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分析实战
扩增子脱靶检测数据分析

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◆前言

on-target编辑效率 / off-target脱靶位置？

预测

（设计序列、参考基因组）

- Off-spotter
- Cas-OFFinder

检测

（设计序列、参考基因组、测序数据）

- Guide-seq (dsODN Tag)
- CRISPResso2

目录

Cas-OFFinder在线网站使用指南

01

Cas-OFFinder

A fast and versatile algorithm that searches for potential off-target sites of Cas9 RNA-guided endonucleases.

http://www.rgenome.net/cas-offinder/

Job title (Optional): 任务名

E-mail (Optional): 邮箱

* The result will be notified by e-mail (*searching job is working in sequence for many input data, therefore it would be convenient to receive the results by e-mail*).

PAM Type PAM序列

CRISPR/Cas-derived RNA-guided Endonucleases (RGENs)

- SpCas9 from Streptococcus pyogenes: 5'-NGG-3'
- SpCas9 from Streptococcus pyogenes: 5'-NRG-3' (R = A or G)
- StCas9 from Streptococcus thermophilus: 5'-NNAGAAW-3' (W = A or T)
- NmCas9 from Neisseria meningitidis: 5'-NNNNGMTT-3' (M = A or C)
- SaCas9 from Staphylococcus aureus: 5'-NNGRRT-'3 (R=A or G)
- CjCas9 from Campylobacter jejuni: 5'-NNNVRYAC-3' (V = G or C or A, R = A or G, Y = C or T)
- CjCas9 from Campylobacter jejuni: 5'-NNNNRYAC-3' (R=A or G, Y= C or T)
- AsCpf1 from Acidaminococcus or LbCpf1 from Lachnospiraceae: 5'-TTTN-3'
- AsCpf1 from Acidaminococcus or LbCpf1 from Lachnospiraceae: 5'-TTTV-3' (V = G or C or A)
- SpCas9 from Streptococcus pasteurianus: 5'-NNGTGA-3'
- FnCpf1 from Francisella: 5'-TTN-3'
- SaCas9 from Staphylococcus aureus: 5'-NNNRRT-'3 (R=A or G)
- FnCpf1 from Francisella: 5'-KYTV-3'
- VRER SpCas9 from Streptococcus pyogenes: 5'-NGCG-3'
- VQR SpCas9 from Streptococcus pyogenes: 5'-NGA-3'

Target Genome 物种

Organism Type

Vertebrates

Genomes

- Homo sapiens (GRCh38/hg38) - Human
- Homo sapiens (hg19) - Human
- Mus musculus (mm10) - Mouse
- Bos taurus (bosTau7) - Cow
- Canis familiaris (canFam3) - Dog
- Rattus norvegicus (rn5) - Rat
- Sus scrofa (susScr11) - Pig
- Danio rerio (danRer7) - Zebrafish
- Macaca mulatta(rheMac3) - Monkey
- Xenopus tropicalis (JGI 4.2/xenTro3) - Western clawed frog

Query Sequences

Query sequences (5' to 3'), one sequence per line.

Please write crRNA sequences **without PAM sequences** (e.g. without NGG for SpCas9).
The length of each query sequence should be between **15** and **25** nt, and **all be the same length**!

CAGCAACTCCAGGGGGCCGC
AAAGGAAACCATTGTGTTAA

等长度sgRNA序列，不包含PAM

Mismatch Number
(eq or less than)
0

DNA Bulge Size
(eq or less than)
0

RNA Bulge Size
(eq or less than)
0

off-target允许的
最大错配、插入、缺失

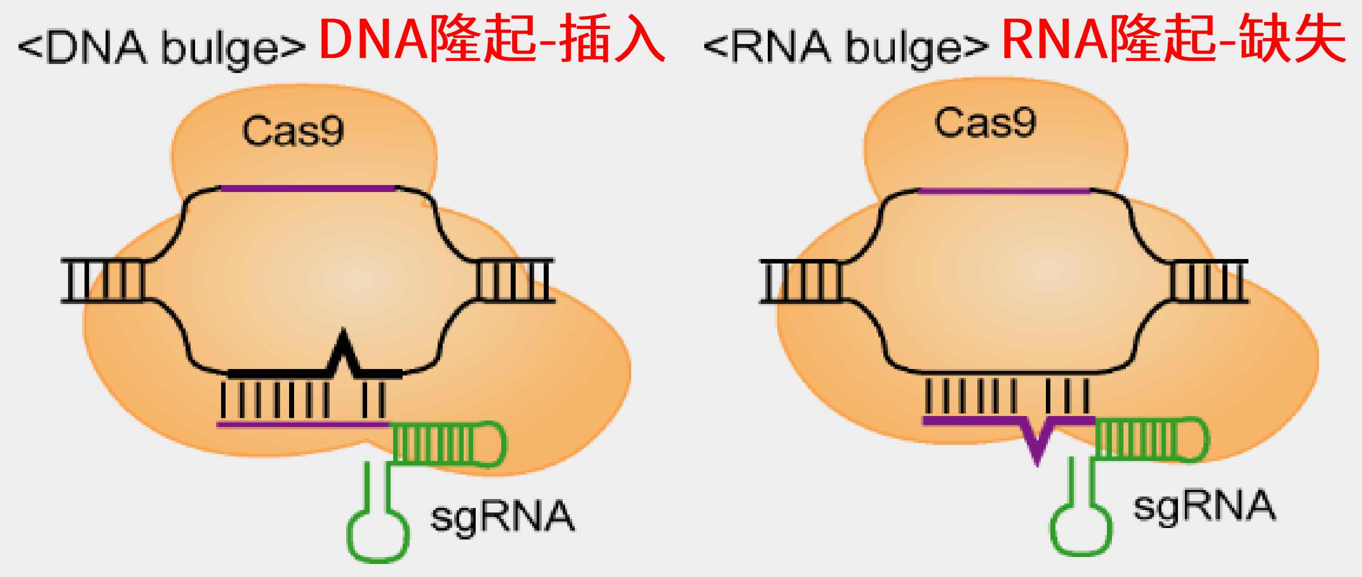
Please note that large number of bulge size will significantly increase the calculation time!

Mixed bases are allowed.

The count of query sequence must be less than 1000.

(In case of NNN and NRN PAM, must be less than 10.)

Submit



❖ Output

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Summary 输入的sgRNA+PAM序列 突起类型 突起长度 错配数 脱靶序列数量

Target Sequence	Bulge Type	Bulge Size	Mismatch	Number of Found Targets
AAAGGAAACCATTGTGTTAANGG	DNA	1	0	3
AAAGGAAACCATTGTGTTAANGG	DNA	1	1	6
AAAGGAAACCATTGTGTTAANGG	DNA	1	2	52
AAAGGAAACCATTGTGTTAANGG	DNA	1	3	632
AAAGGAAACCATTGTGTTAANGG	DNA	2	0	3
AAAGGAAACCATTGTGTTAANGG	DNA	2	1	2
AAAGGAAACCATTGTGTTAANGG	DNA	2	2	26
AAAGGAAACCATTGTGTTAANGG	DNA	2	3	569

Details

过滤与下载

Bulge Type | DNA bulge | RNA bulge | Mismatch | Filter | Download filtered result

On/Off target比对 Off target位置

Bulge Type	Target	Chromosome	Position	Direction	Mismatches	Bulge Size
X	crRNA: CAGCAACTCCAGGGGGCCGCNGG DNA: CAGCAACTCCAGaaGGCCGCTGG	chr8	37610874	-	2	0
DNA	crRNA: C-AGCAACTCCAGGGGGCCGCNGG DNA: CCAGCAACTCCAGaaGGCCGCTGG	chr8	37610874	-	2	1
DNA	crRNA: CA-GCAACTCCAGGGGGCCGCNGG DNA: CcAGCAACTCCAGaaGGCCGCTGG	chr8	37610874	-	3	1
DNA	crRNA: CAG-CAACTCCAGGGGGCCGCNGG DNA: CAGACcACTCCAGGGaGCtGCCGG	chr8	1540629	+	3	1
DNA	crRNA: CAGCAACTCCAG-GGGGCCGCNGG DNA: CAGCcACTtCAGAGGGGCtGCAGG	chr8	11566198	-	3	1

CRISPResso2在线网站使用指南

02

CRISPResso2

Analysis of genome editing outcomes
from deep sequencing data

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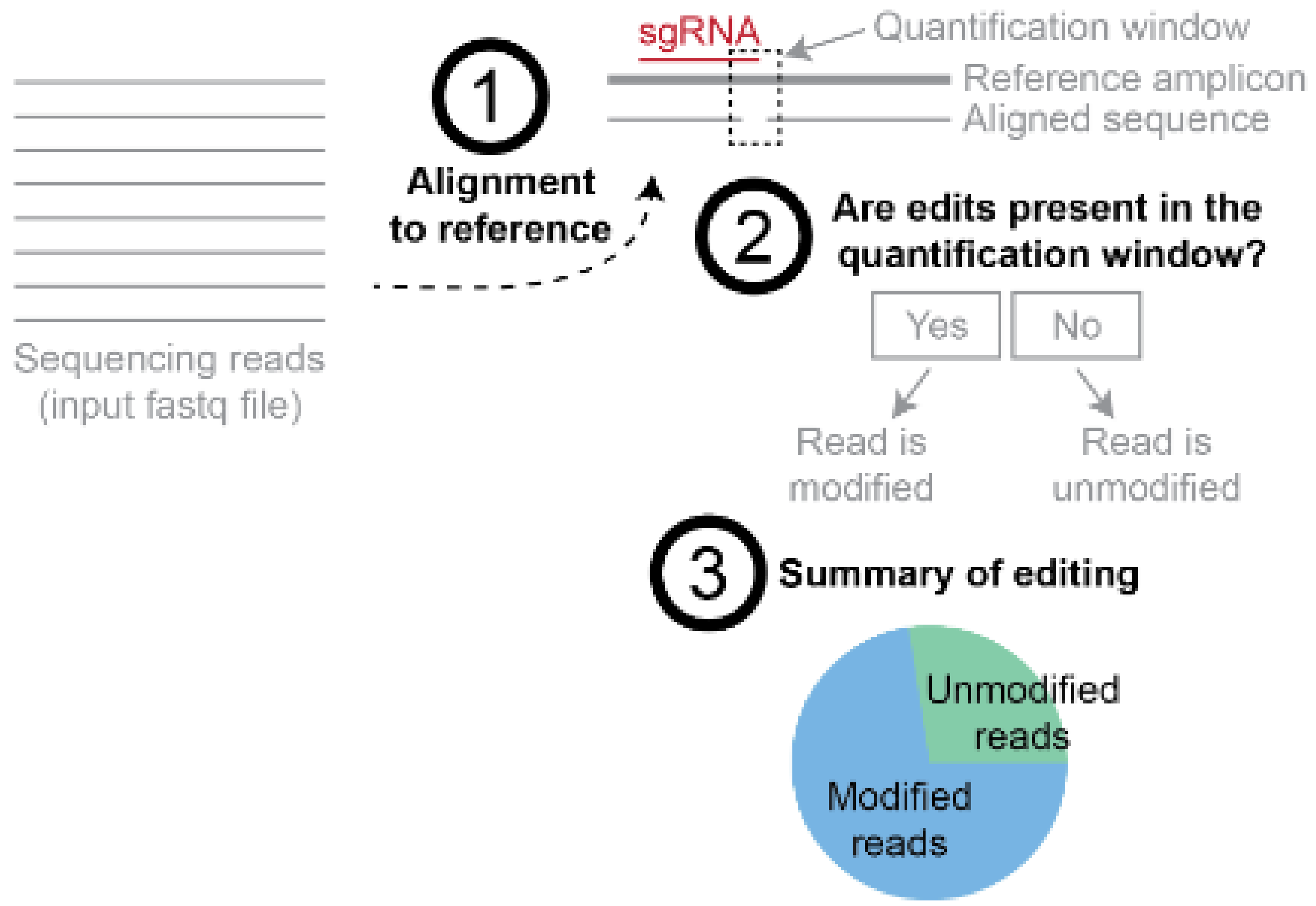
http://crispresso.pinellolab.org/submission

➢ 是什么?

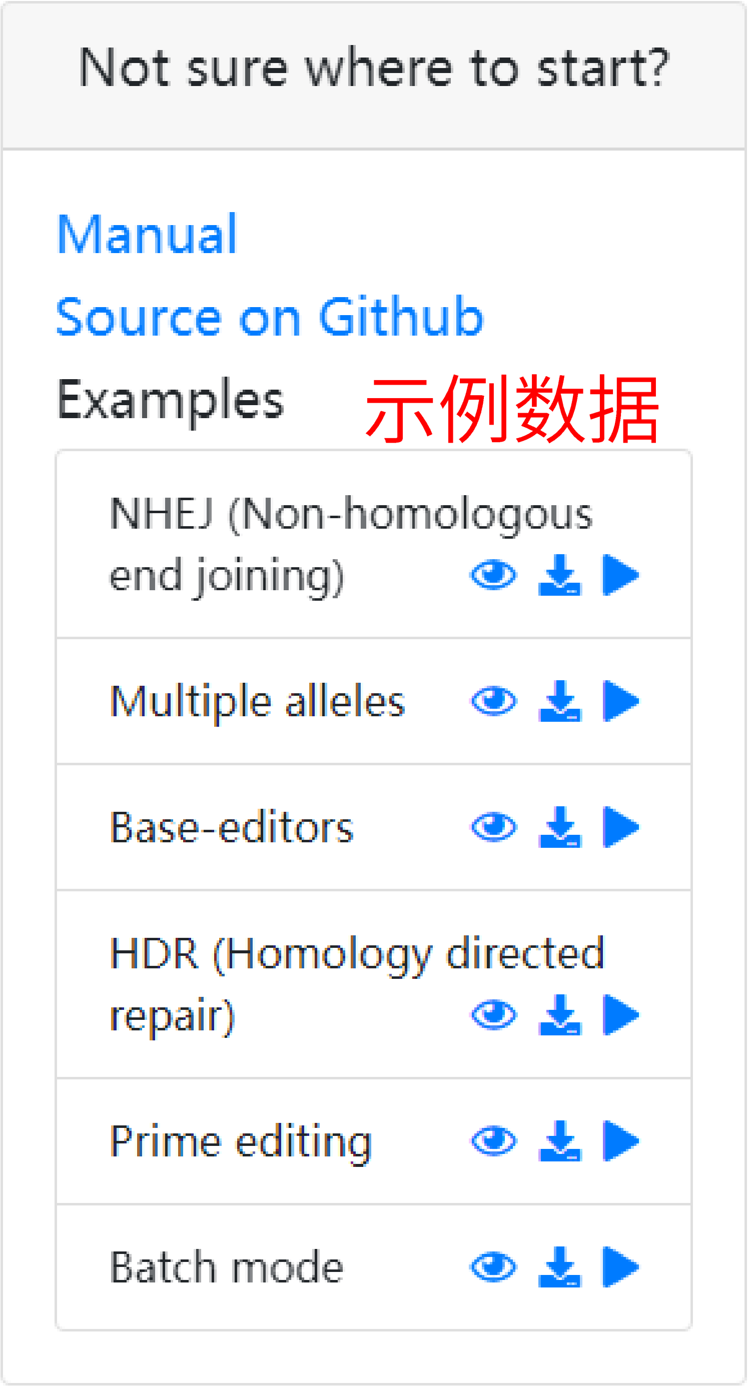
分析基因组编辑实验的软件流程，实现对扩增子测序产生的结果进行快速直观的解读。

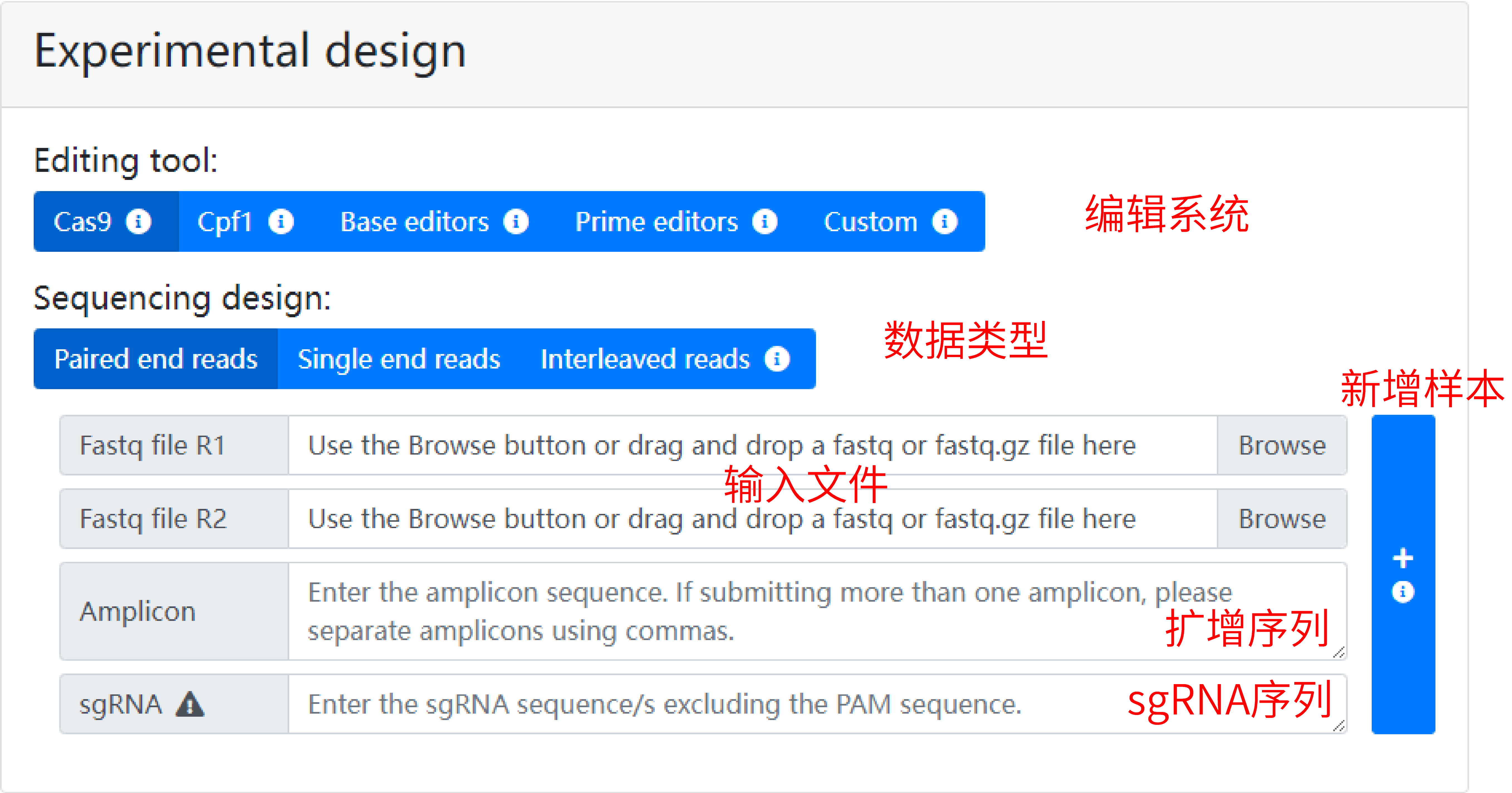
➢ 做什么?

- Alignment：测序reads与参考序列比对
- Quantization：量化插入、突变和缺失，确定reads是否被编辑
- Visualization：以直观的图表和数据集总结编辑结果

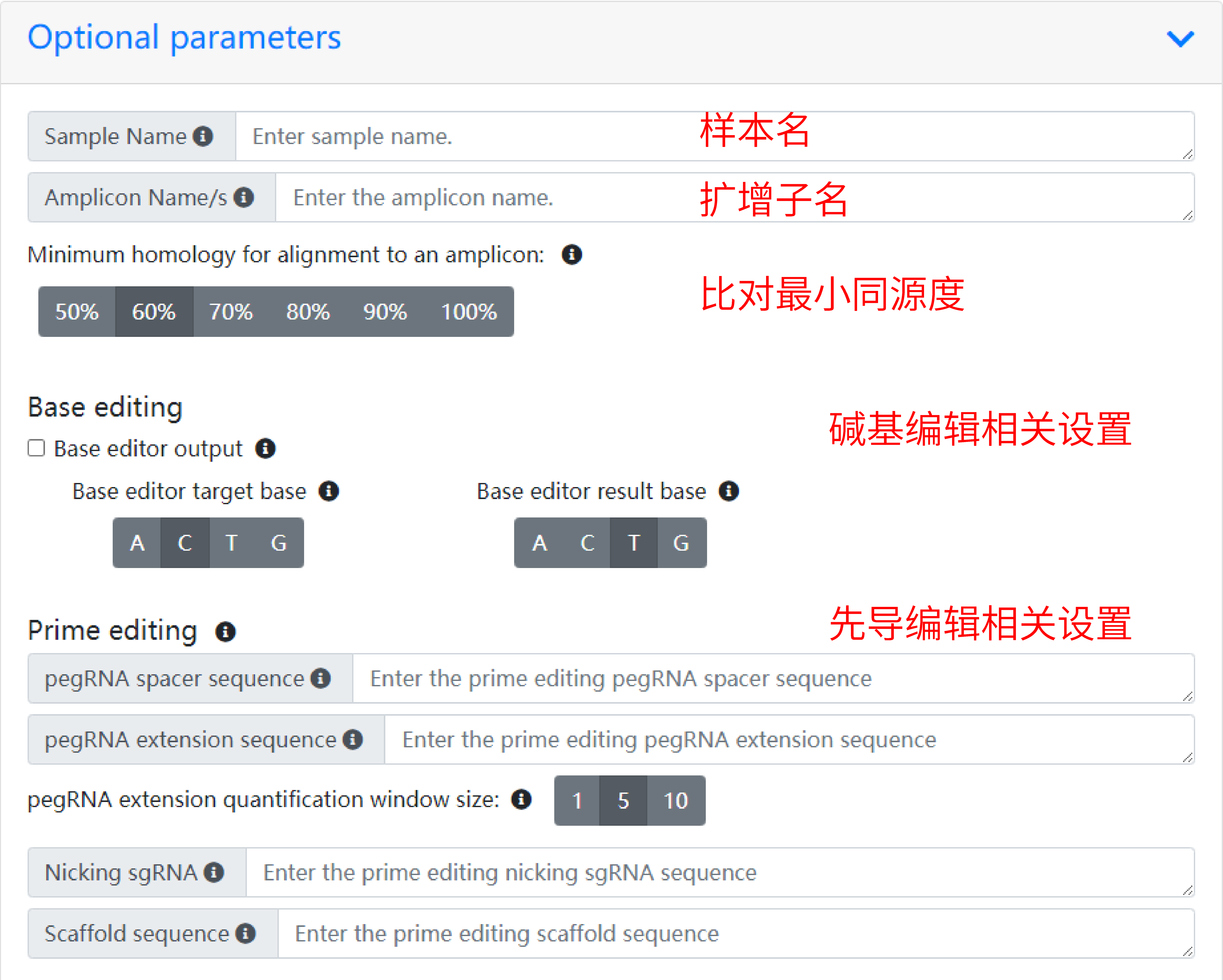


➢ 实验设计


Not sure where to start?
Manual
Source on Github
Examples
示例数据
NHEJ (Non-homologous end joining)
Multiple alleles
Base-editors
HDR (Homology directed repair)
Prime editing
Batch mode


Experimental design
Editing tool:
Cas9
Cpf1
Base editors
Prime editors
Custom
编辑系统
Sequencing design:
Paired end reads
Single end reads
Interleaved reads
数据类型
新增样本
Fastq file R1
Use the Browse button or drag and drop a fastq or fastq.gz file here
Browse
输入文件
Fastq file R2
Use the Browse button or drag and drop a fastq or fastq.gz file here
Browse
Amplicon
Enter the amplicon sequence. If submitting more than one amplicon, please separate amplicons using commas.
扩增序列
sgRNA
Enter the sgRNA sequence/s excluding the PAM sequence.
sgRNA序列

➢ 参数设置


Optional parameters
Sample Name
Enter sample name.
样本名
Amplicon Name/s
Enter the amplicon name.
扩增子名
Minimum homology for alignment to an amplicon:
50% 60% 70% 80% 90% 100%
比对最小同源度
Base editing
Base editor output
碱基编辑相关设置
Base editor target base
A C T G
Base editor result base
A C T G
Prime editing
先导编辑相关设置
pegRNA spacer sequence
Enter the prime editing pegRNA spacer sequence
pegRNA extension sequence
Enter the prime editing pegRNA extension sequence
pegRNA extension quantification window size:
1 5 10
Nicking sgRNA
Enter the prime editing nicking sgRNA sequence
Scaffold sequence
Enter the prime editing scaffold sequence


Quantification window
Center of the quantification window (relative to 3' end of the provided sgRNA):
-15 -10 -3 0 +1
定量窗口中心
Quantification window size (bp):
定量窗口大小
No window 1 2 3 4 5 6 7 8 9 10 15 20 25 30
Plot window size (bp):
5 10 20 30
绘图窗口大小
Ignore substitutions
Ignore insertions
Ignore deletions
忽略碱基替换、插入、缺失
HDR
Expected HDR amplicon sequence:
HDR扩增序列
Enter the amplicon sequence expected after HDR with the donor sequence
Exon specification
编码序列
Coding Sequence/s:
Enter the subsequence/s of the amplicon sequence comprising coding sequences for frameshift analysis. If entering more than one coding exon, please separate sequences using commas

➢ 参数设置

Quality filtering and trimming

Minimum average read quality (phred33 scale):

No Filter | >10 | >20 | >30 | >35

Read平均质量值

Minimum single bp quality (phred33 scale):

No Filter | >10 | >20 | >30 | >35

单碱基质量值

Replace bases with N that have a quality lower than (phred33 scale):

No Filter | <10 | <20 | <30 | <35

替换为N碱基

Exclude bp from the left side of the amplicon sequence for the quantification of the mutations:

Disabled | 5 | 10 | 15 | 20 | 40 | 50

忽略扩增子左侧碱基

Exclude bp from the right side of the amplicon sequence for the quantification of the mutations:

Disabled | 5 | 10 | 15 | 20 | 40 | 50

忽略扩增子右侧碱基

Trimming adapter:

No Trimming | Nextera PE | TruSeq3 PE | TruSeq3 SE | TruSeq2 PE | TruSeq2 SE

去除接头类型

Email

邮箱

By clicking the submit button, I agree to use CRISPResso2 in accordance with Partners' Terms and Conditions.

Submit!

接头类型?

➢ 结果解读

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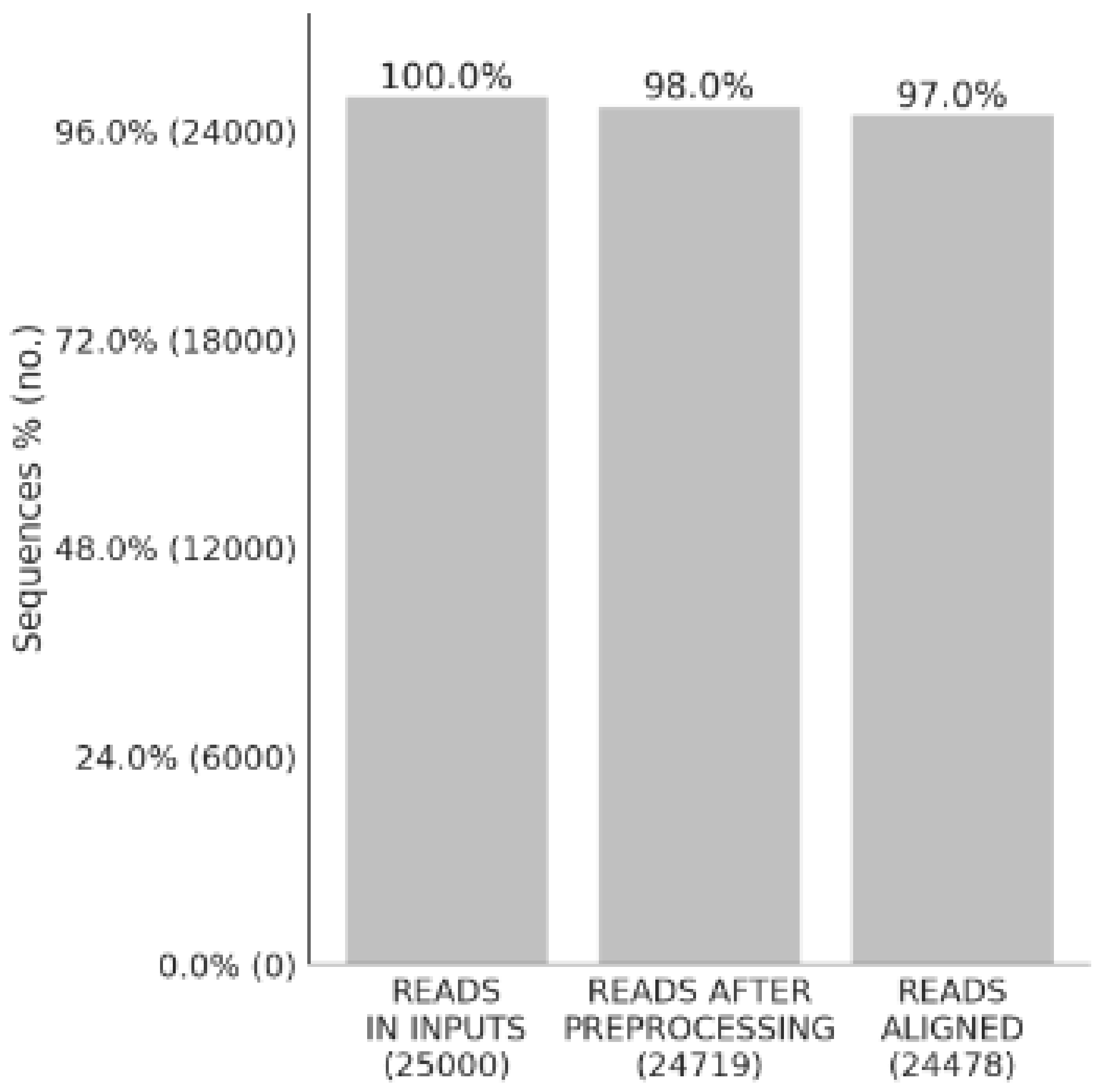

图1a 质控、比对Reads统计

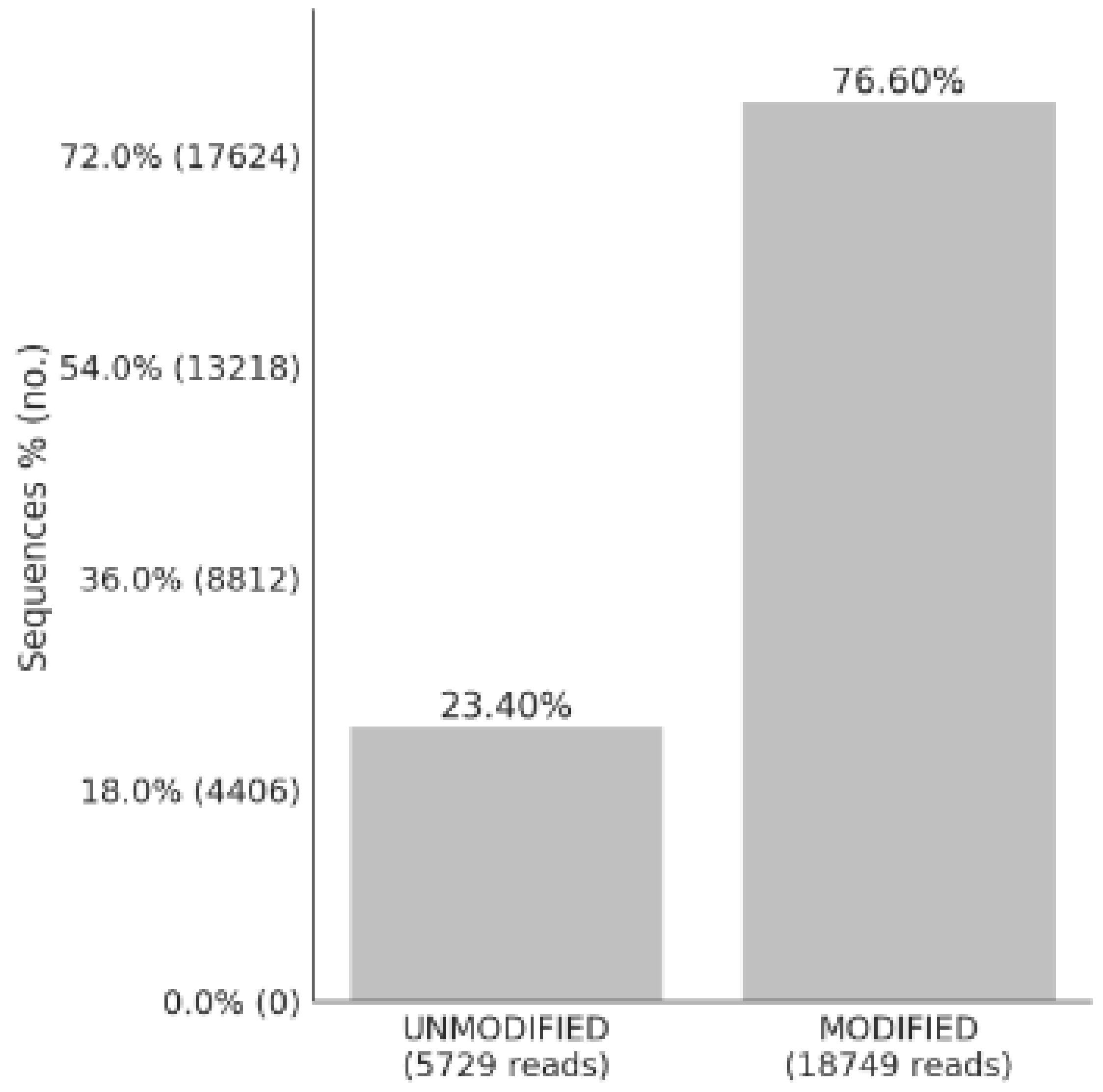

图1b 编辑Reads统计

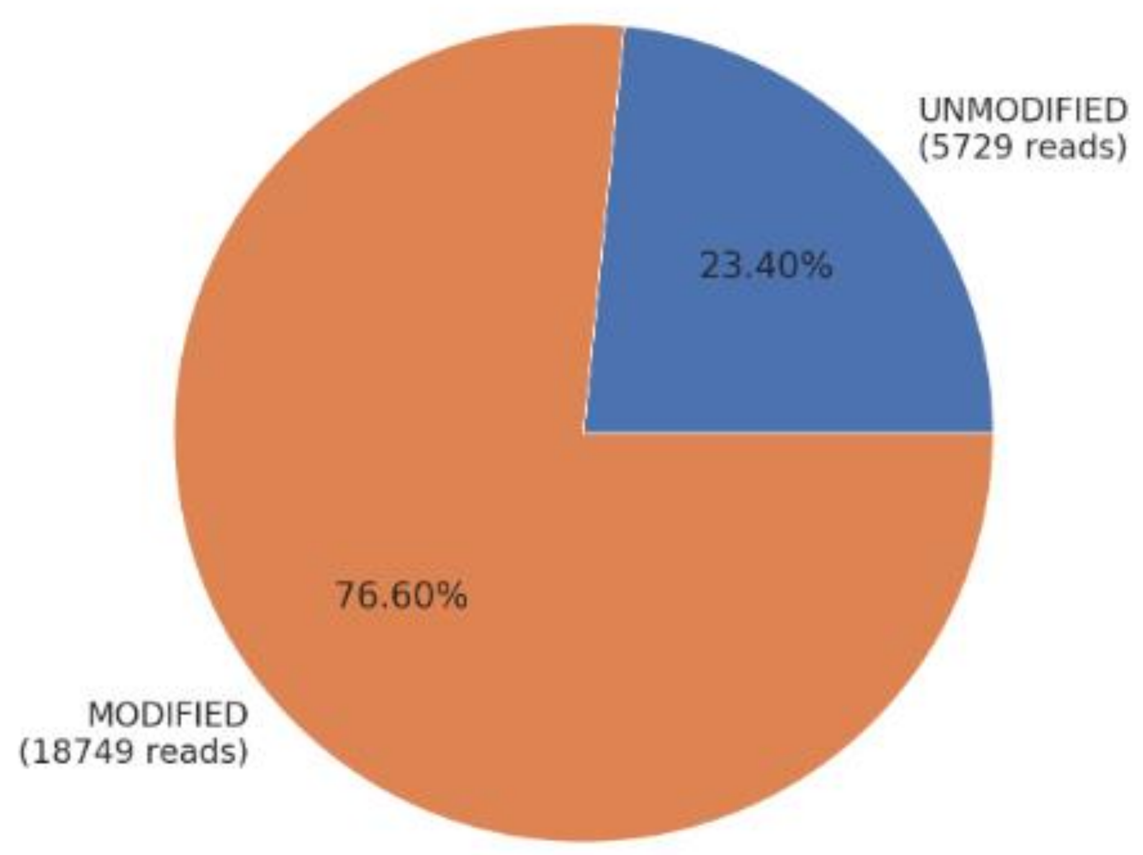

图1c 编辑Reads统计

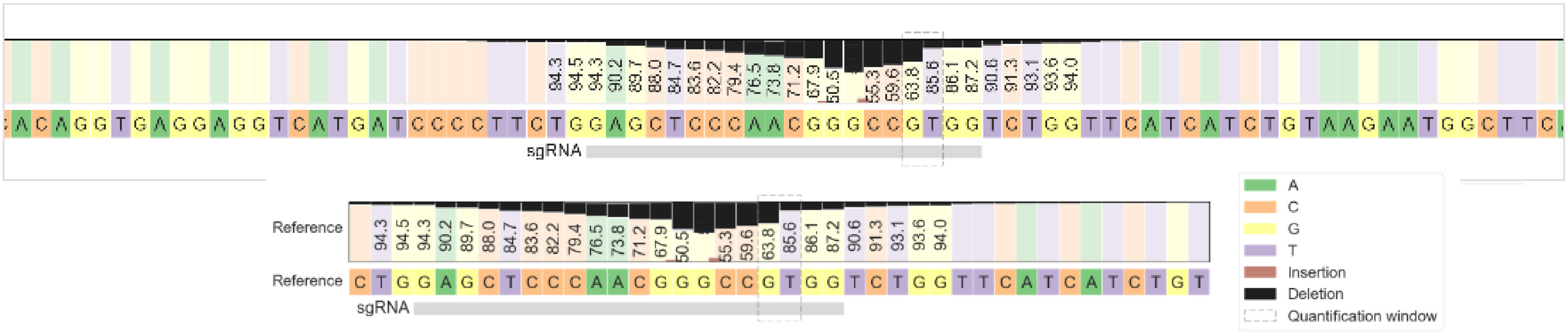

图2a&2b 核苷酸分布

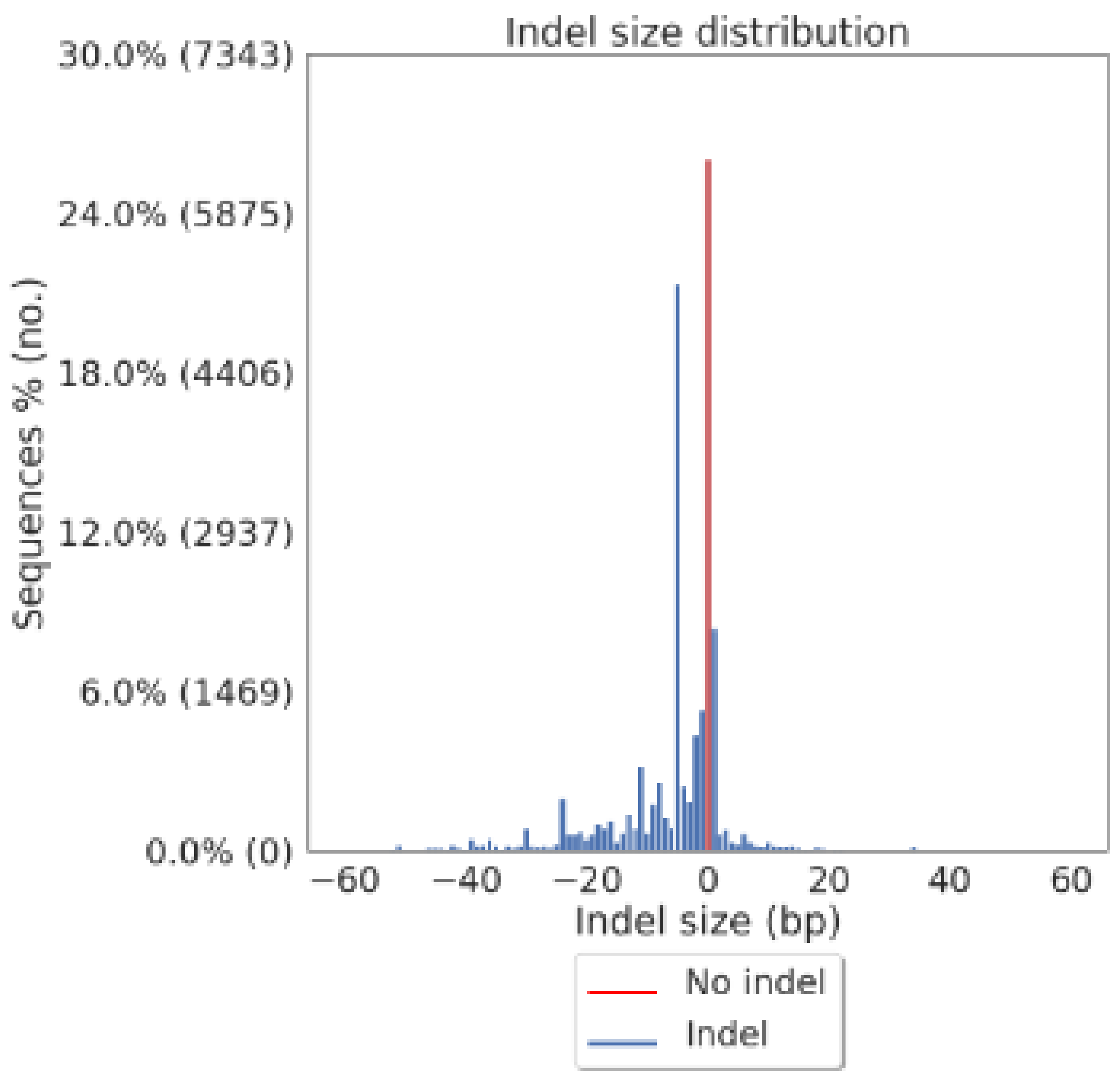

图3a InDel统计

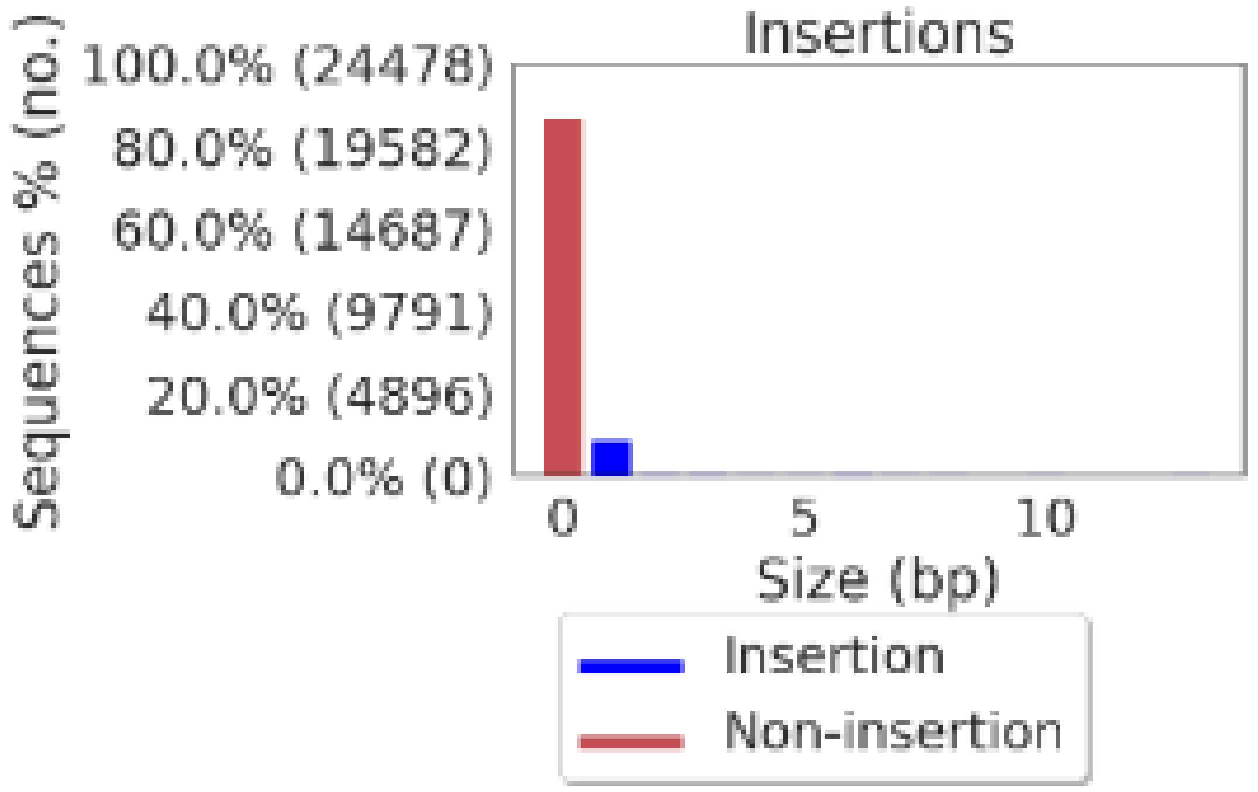

图3b 插入统计

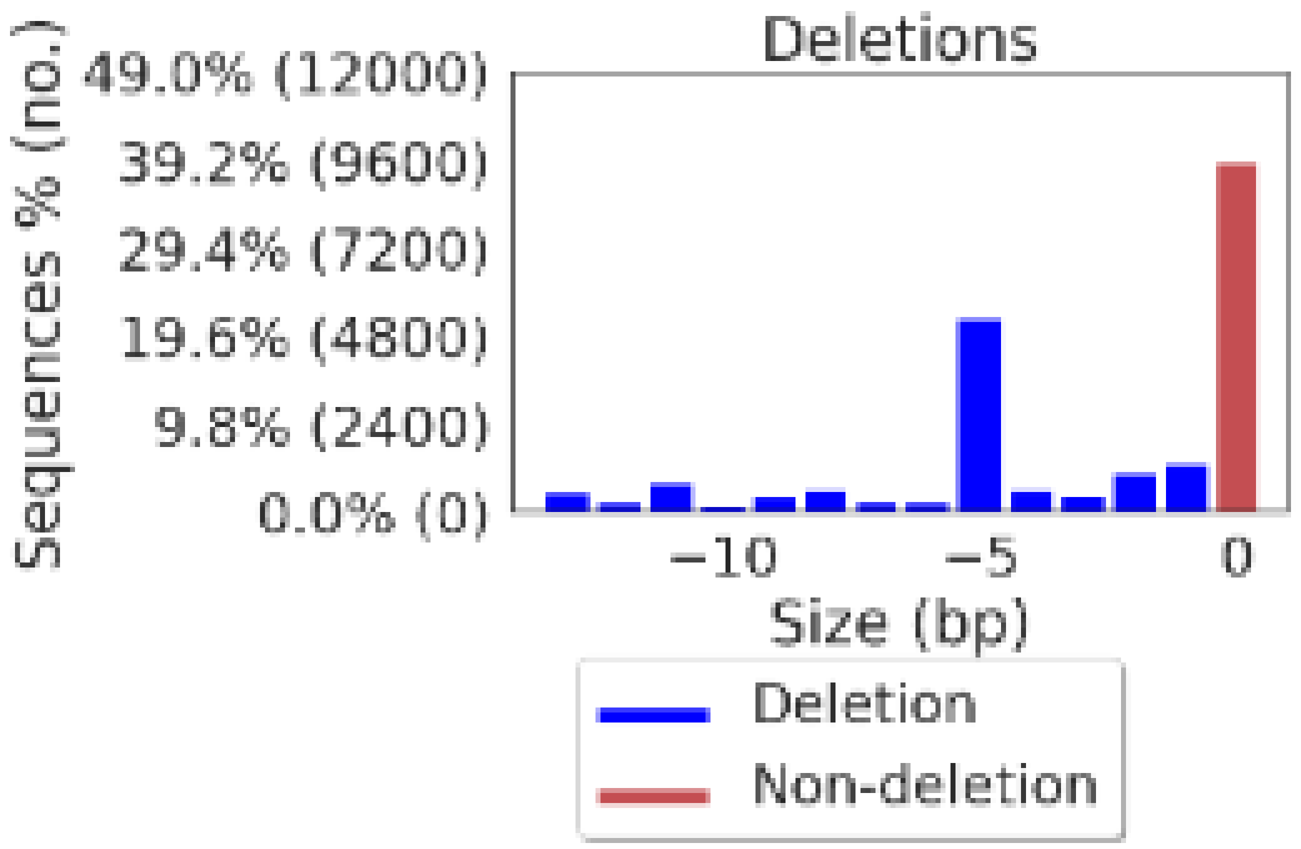

图3b 缺失统计

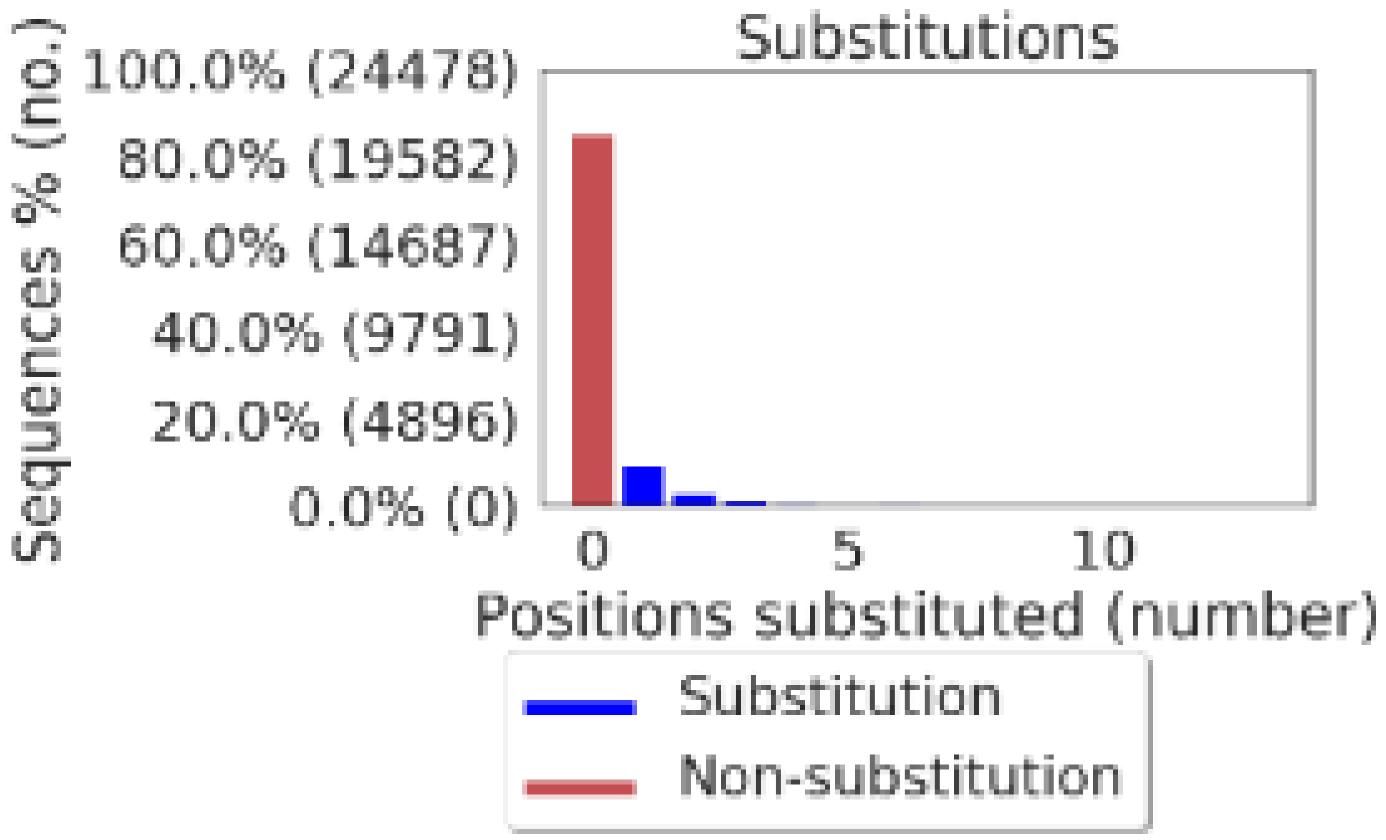

图3b 碱基替换统计

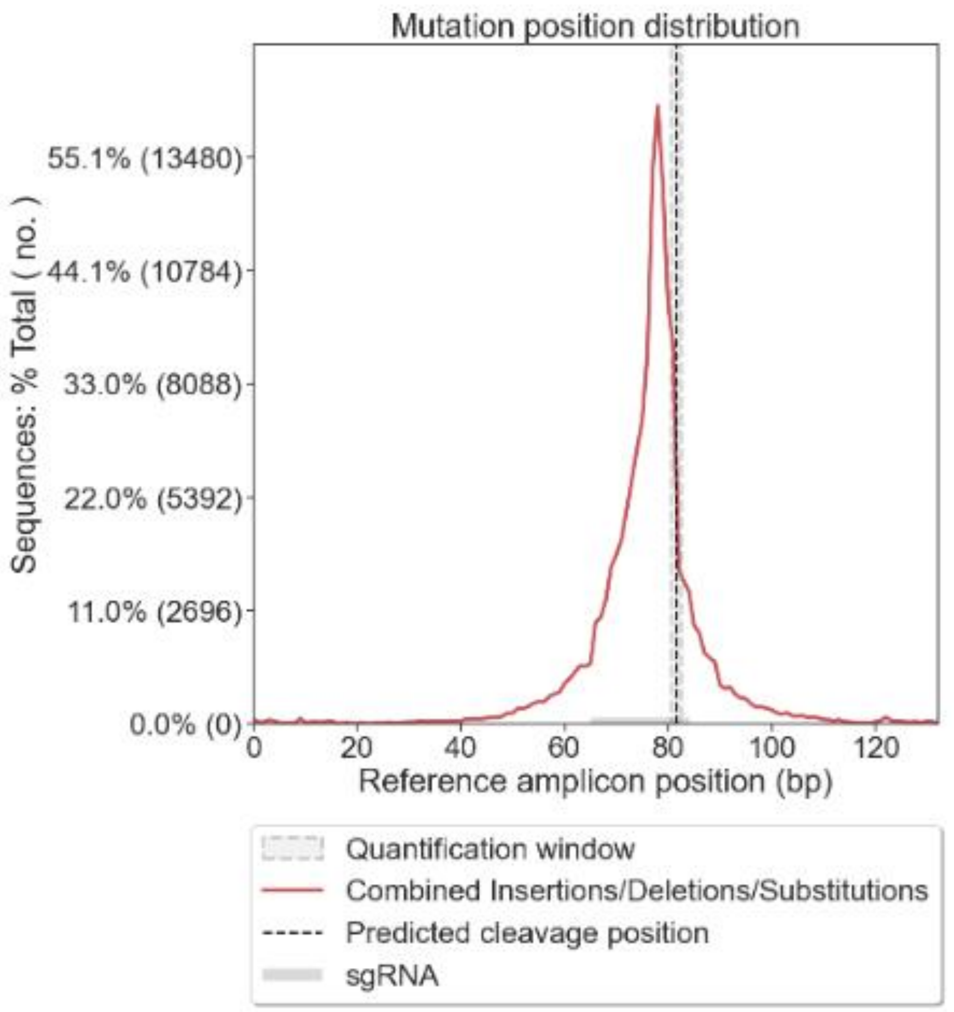

图4a 扩增子突变统计

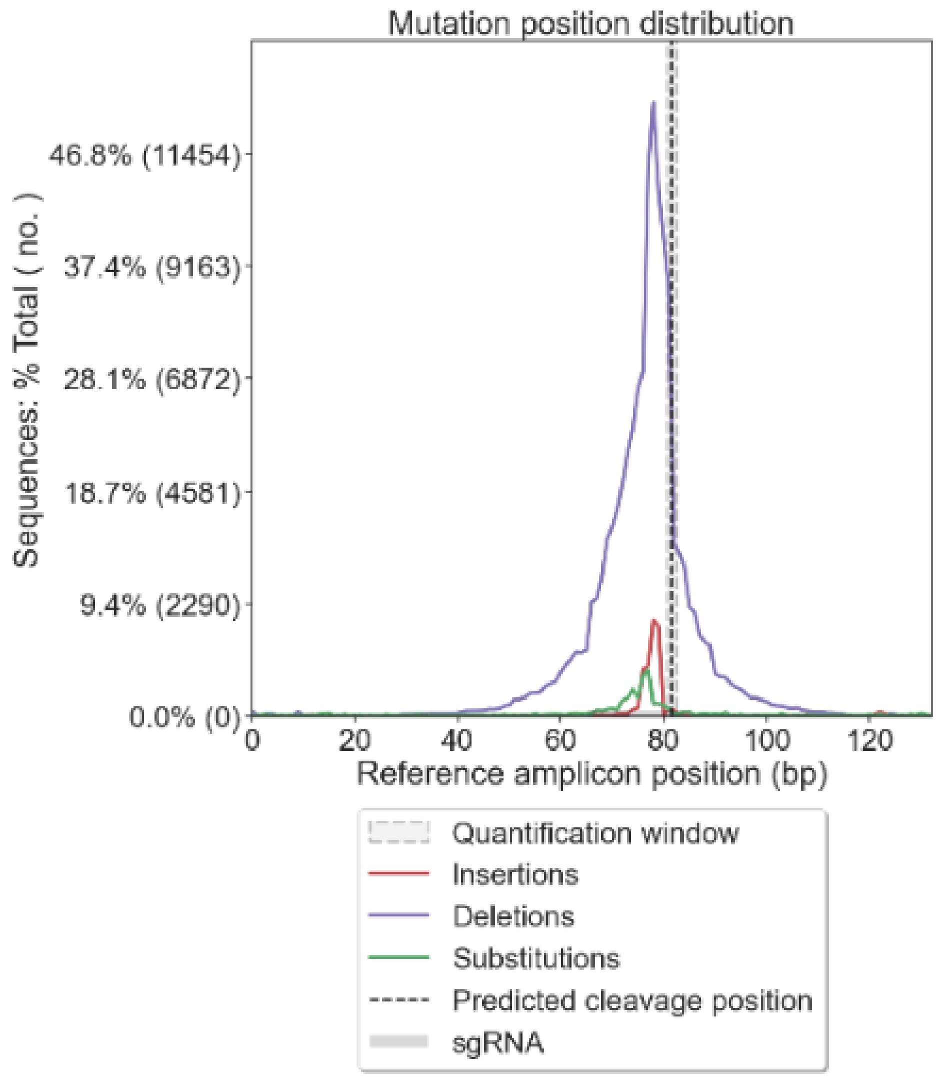

图4b 扩增子插入、缺失、替换统计

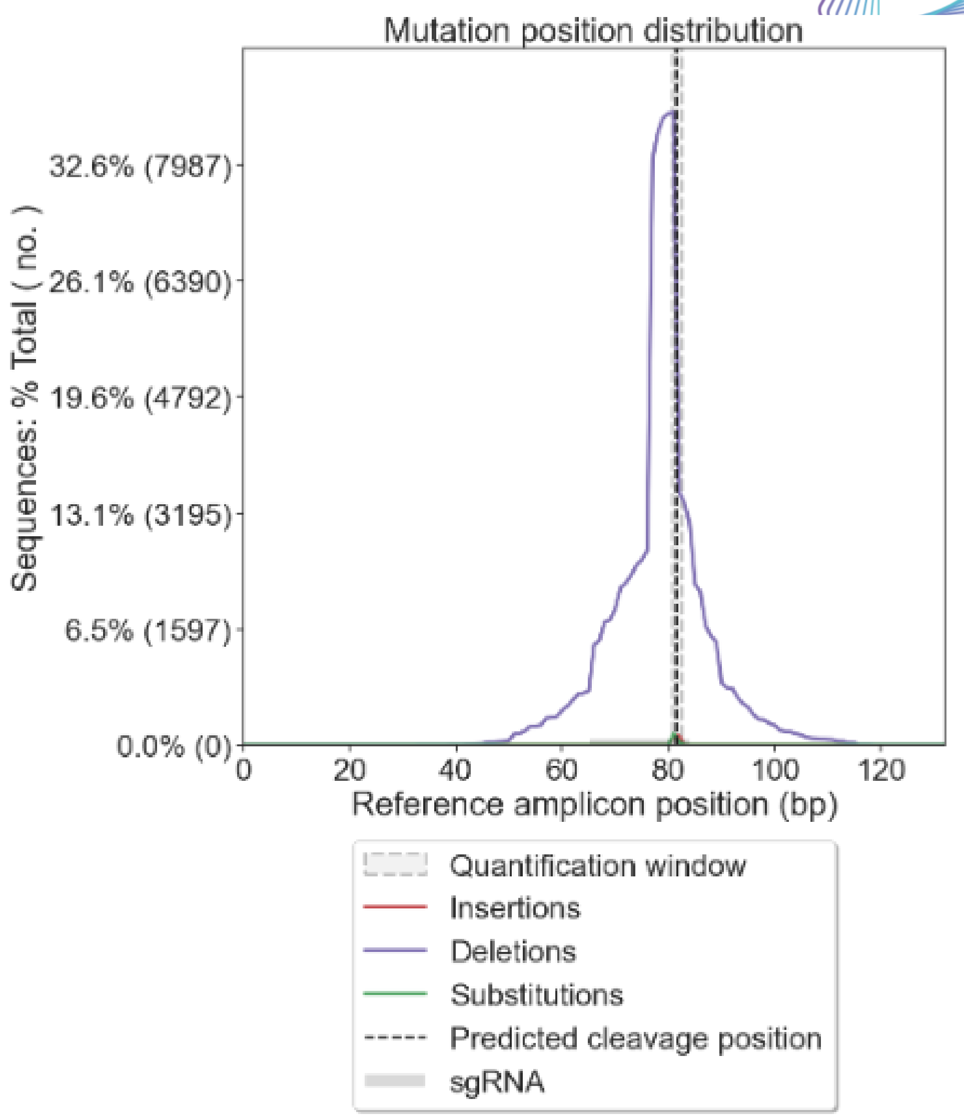

图4c 与定量窗口重叠的插入、缺失、替换统计

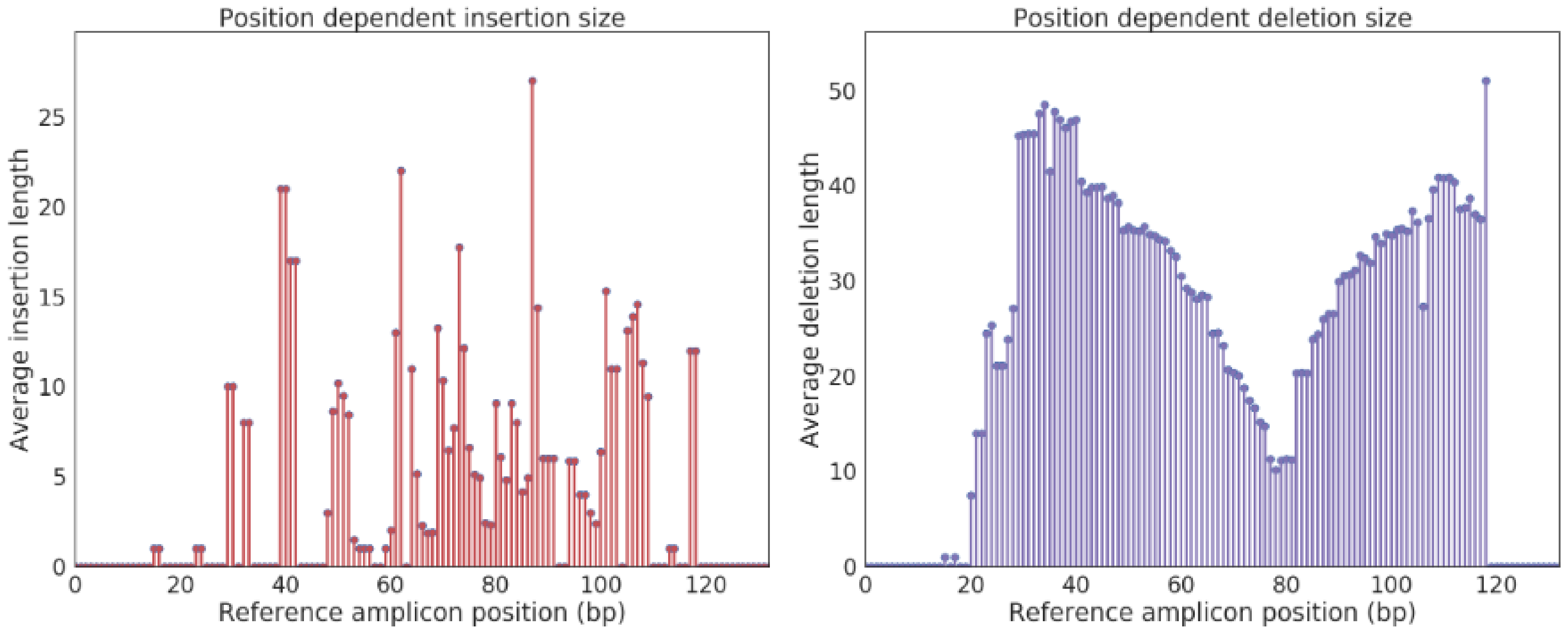

图4d 与定量窗口重叠的InDel长度统计

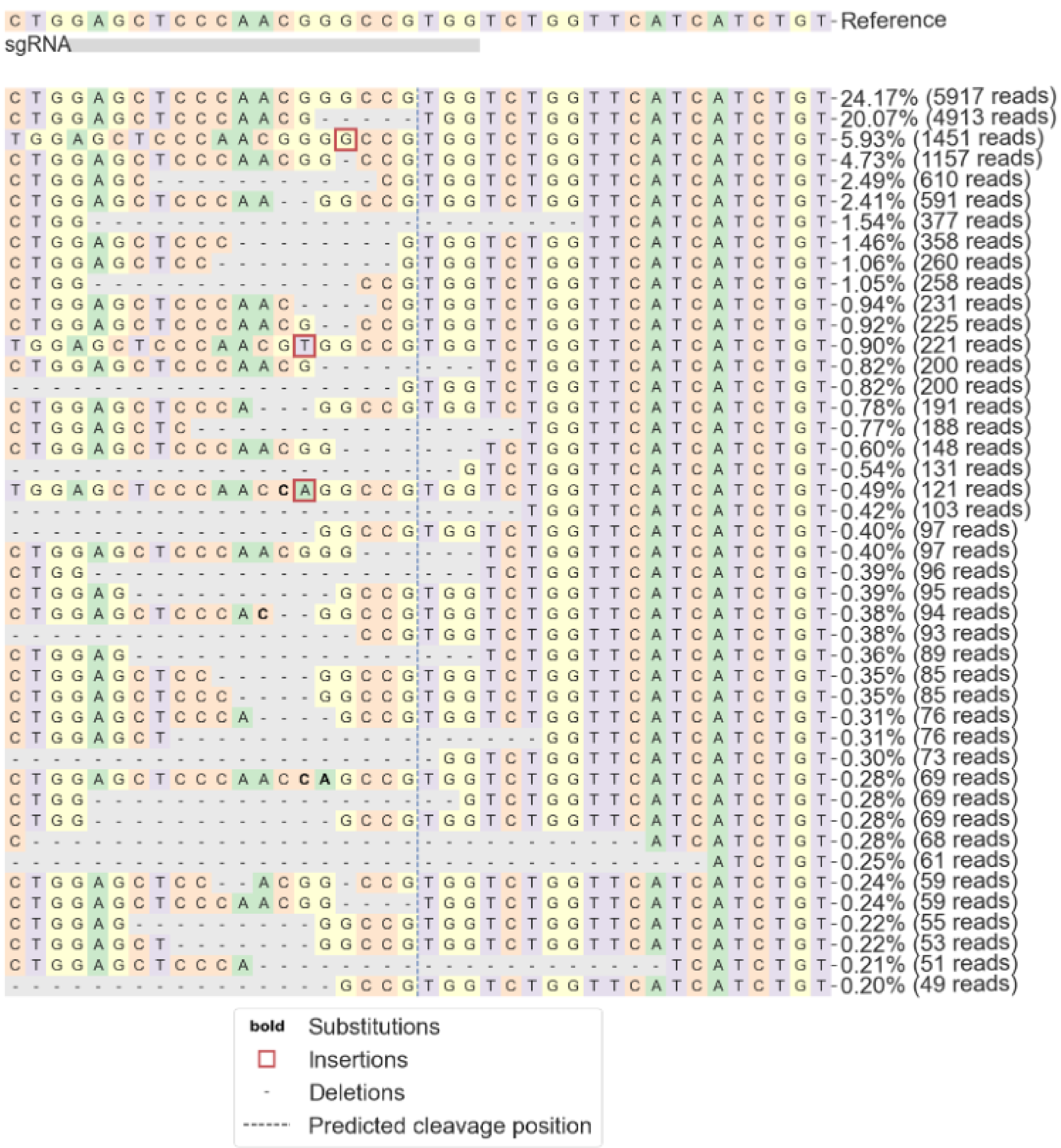

图9. 序列类型可视化

数据量太大？
参数设置不自由？

CRISPResso2本地流程使用指南

03

所需软件列表：

- Cutadapt https://cutadapt.readthedocs.io/en/stable/installation.html
- CRISPResso2 https://github.com/pinellolab/CRISPResso2

推荐使用Conda安装：

```
#Cutadapt
conda config --add channels bioconda
conda config --add channels conda-forge
conda config --set channel_priority strict
conda create -n cutadapt cutadapt
```

```
#CRISPResso2
conda config --add channels defaults
conda config --add channels bioconda
conda config --add channels conda-forge
conda install CRISPResso2
```

➢ Step1 数据质控：对测序数据进行过滤，去除接头序列、低质量序列与短序列

cutadapt –h

- 常用参数：

-a/--adapter	read1 3’接头序列
-g/--front	read1 5’接头序列
-A	read2 3’接头序列
-G	read2 5’接头序列
-e/--error-rate	接头匹配错误率，默认0.1
-q/--quality-cutoff	过滤质量值，平均值低于此值的reads被丢弃
-m/--minimum-length	过滤长度，低于此值的reads被丢弃
--max-n	N碱基数量/含量，高于此值的reads被丢弃
-o	read1输出文件
-p	read2输出文件

- e.g.

cutadapt -a AGATCGGAAGAGCACACGTCTGAACTCCAGTCA -A AGATCGGAAGAGCGTCGTGTAGGGAAAGAGTGT -o Sample.R1.trimed.fq.gz -p Sample.R2.trimed.fq.gz /CRISPR/CRISPResso/Sample.R1.fq.gz /CRISPR/CRISPResso/Sample.R2.fq.gz

➢ Step2 脱靶分析

CRISPResso -h

- 常用参数：

-r1/--fastq_r1	read1 fastq格式文件
-r2/--fastq_r2	read2 fastq格式文件
-a/--amplicon_seq	扩增子序列
-an/--amplicon_name	扩增子名称
--min_identity_score	最小同源度
-g/--guide_seq/--sgRNA	sgRNA序列
-gn/--guide_name	sgRNA名称
-q/--min_average_read_quality	最低平均质量分数
-s/--min_single_bp_quality	最低单个碱基质量分数
--file_prefix	输出图表和表格的文件前缀
-n/--name	报告的输出名称
-o/--output_folder	输出目录
-w/--quantification_window_size/--window_around_sgrna	量化窗口大小
-wc/--quantification_window_center/--cleavage_offset	量化窗口中心位置（Cas9系统: -3）
--ignore_substitutions	忽略替换事件
--ignore_insertions	忽略插入事件
--ignore_deletions	忽略缺失事件
--plot_window_size/--offset_around_cut_to_plot	从量化窗口中心延伸出来绘图窗口大小

- e.g.

CRISPResso -r1 Sample.R1.trimed.fq.gz **-r2** Sample.R2.trimed.fq.gz

-o output_dir **–sgRNA** AAAGGCTGCTGATGACACCT

–amplicon_seq AGTGCTGAAGTGATTACGGTGTGCGGGCACCGGTGAATCCA

AGTGTCCTCTGATGGTCAAAGTTCTAGATGCTGTCCGAGGCAGTCCTGCCATCAA

TGTGGCCGTGCATGTGTTCAGAAAGGCTGCTGATGACACCTGGGAGCCATTTGC

CTCTGGGTAAGTTGCCAAAGAACCCTCCCACAGGACTTGGTTTTATCTTCCCGTT

TGCCCCTCACTTGGTAGAGAGAGGCTCACCATCCAGCATCCAACTCTCCGGA

```
1a.Read_barplot.pdf
1a.Read_barplot.png
1b.Alignment_pie_chart.pdf
1b.Alignment_pie_chart.png
1c.Alignment_barplot.pdf
1c.Alignment_barplot.png
2a.Nucleotide_percentage_quilt.pdf
2a.Nucleotide_percentage_quilt.png
2b.Nucleotide_percentage_quilt_around_sgRNA_AAAGGCTGCTGATGACACCT.pdf
2b.Nucleotide_percentage_quilt_around_sgRNA_AAAGGCTGCTGATGACACCT.png
3a.Indel_size_distribution.pdf
3a.Indel_size_distribution.png
3b.Insertion_deletion_substitutions_size_hist.pdf
3b.Insertion_deletion_substitutions_size_hist.png
4a.Combined_insertion_deletion_substitution_locations.pdf
4a.Combined_insertion_deletion_substitution_locations.png
4b.Insertion_deletion_substitution_locations.pdf
4b.Insertion_deletion_substitution_locations.png
4c.Quantification_window_insertion_deletion_substitution_locations.pdf
4c.Quantification_window_insertion_deletion_substitution_locations.png
4d.Position_dependent_average_indel_size.pdf
4d.Position_dependent_average_indel_size.png
9.Alleles_frequency_table_around_sgRNA_AAAGGCTGCTGATGACACCT.pdf
9.Alleles_frequency_table_around_sgRNA_AAAGGCTGCTGATGACACCT.png
Alleles_frequency_table_around_sgRNA_AAAGGCTGCTGATGACACCT.txt
Alleles_frequency_table.zip
CRISPResso2_info.pickle
CRISPResso_mapping_statistics.txt
CRISPResso_quantification_of_editing_frequency.txt
CRISPResso_RUNNING_LOG.txt
Deletion_histogram.txt
Effect_vector_combined.txt
Effect_vector_deletion.txt
Effect_vector_insertion.txt
Effect_vector_substitution.txt
Indel_histogram.txt
Insertion_histogram.txt
Modification_count_vectors.txt
Nucleotide_frequency_table.txt
Nucleotide_percentage_table.txt
Quantification_window_modification_count_vectors.txt
Quantification_window_nucleotide_frequency_table.txt
Quantification_window_nucleotide_percentage_table.txt
Substitution_histogram.txt
```

- **Output**

CRISPResso_on_Sample.R1_Sample.R2/（结果目录）

CRISPResso_on_Sample.R1_Sample.R2.html（网页报告）

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问题解答

04

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References

[1] Martin, Marcel. "Cutadapt removes adapter sequences from high-throughput sequencing reads." EMBnet. journal 17.1 (2011): pp-10.

[2] Clement K, Rees H, Canver MC, Gehrke JM, Farouni R, Hsu JY, Cole MA, Liu DR, Joung JK, Bauer DE, Pinello L. CRISPResso2 provides accurate and rapid genome editing sequence analysis. Nat Biotechnol. 2019 Mar; 37(3):224-226. doi: 10.1038/s41587-019-0032-3. PubMed PMID: 30809026.

[3] Bae S., Park J., & Kim J.-S. Cas-OFFinder: A fast and versatile algorithm that searches for potential off-target sites of Cas9 RNA-guided endonucleases. Bioinformatics 30, 1473-1475 (2014).

Thank you!

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